Generator (g): 24 Another Generator *Chooses a primitive root modulo p*

> ***Important:*** *Both Alice and Bob can see the same public values (p, g). These are not secret and can be transmitted over an insecure channel.*

Step 2: Private Key Generation and Public Key Calculation

> *Each participant generates their own private key (secret) and calculates their public key using the formula:* ***Public Key = g^(private key) mod p***

Alice

Private Key (a):

5665 Generate A

Alice's secret private key (kept secret)

Public Key (g^a mod p):

2787 Calculate g^a

Alice's public key (can be shared openly)

Send Public Key to Bob

Received from Bob:

3459

Bob's public key (g^b mod p)

Shared Secret:

Calculate Shared Secret

6029

Final shared secret: (g^b)^a mod p = g^(ab) mod p

Bob

Private Key (b):

2131 Generate B

Bob's secret private key (kept secret)

Public Key (g^b mod p):

3459 Calculate g^b

Bob's public key (can be shared openly)

Send Public Key to Alice

Received from Alice:

2787

Alice's public key (g^a mod p)

Shared Secret:

Calculate Shared Secret

6029

Final shared secret: (g^a)^b mod p = g^(ab) mod p

Step 3: Verification

> ***Success!*** *If both Alice and Bob calculated the same shared secret value, they have successfully established a secure communication channel. This shared secret can now be used as an encryption key for further communication.*

> ***Security Note:*** *Even though an eavesdropper can see all the public information (p, g, g^a mod p, g^b mod p), they cannot easily compute the shared secret g^(ab) mod p without solving the discrete logarithm problem, which is computationally difficult for large numbers.*